

1. Closed-Ended Question:

What are the top 5 brands by receipts scanned among users 21 and over?

Assumptions:

- A user's age is determined using their BIRTH_DATE.
- "Receipts scanned" refers to the number of transactions recorded per brand.
- Only users aged **21 and over** are included.

Query:

```
WITH AgeFilteredUsers AS (  
    SELECT ID  
    FROM Users  
    WHERE TIMESTAMPDIFF(YEAR, BIRTH_DATE, CURDATE()) >= 21  
)  
SELECT  
    p.BRAND,  
    COUNT(t.RECEIPT_ID) AS receipts_scanned  
FROM Transactions t  
JOIN Products p ON t.BARCODE = p.BARCODE  
JOIN AgeFilteredUsers u ON t.USER_ID = u.ID  
GROUP BY p.BRAND  
ORDER BY receipts_scanned DESC  
LIMIT 5;
```

What are the top 5 brands by sales among users that have had their account for at least six months?

Assumptions:

- A user's account age is determined by the difference between CREATED_DATE and the current date (CURDATE()).

- Sales are determined by the FINAL_SALE column in the Transactions table.
- The query filters users whose accounts have been active for **at least six months**.

SQL query:

```
WITH EligibleUsers AS (
    SELECT ID
    FROM Users
    WHERE TIMESTAMPDIFF(MONTH, CREATED_DATE, CURDATE()) >= 6
)
SELECT
    p.BRAND,
    SUM(t.FINAL_SALE) AS total_sales
FROM Transactions t
JOIN Products p ON t.BARCODE = p.BARCODE
JOIN EligibleUsers u ON t.USER_ID = u.ID
GROUP BY p.BRAND
ORDER BY total_sales DESC
LIMIT 5;
```

2. Open-Ended Question:

Who are Fetch's power users?

Assumptions:

- A "power user" is defined as a user who has scanned **at least 50 receipts**.
- Users who have made **high-value purchases** (total sales over \$500) are also considered power users.

SQL query:

```
SELECT
    u.ID,
```

```
COUNT(t.RECEIPT_ID) AS total_receipts,  
SUM(t.FINAL_SALE) AS total_spent  
FROM Users u  
JOIN Transactions t ON u.ID = t.USER_ID  
GROUP BY u.ID  
HAVING total_receipts >= 50 OR total_spent > 500  
ORDER BY total_receipts DESC, total_spent DESC;
```

Which is the leading brand in the Dips & Salsa category?

Assumptions:

- The leading brand is determined based on **total sales** in the "Dips & Salsa" category.
- "CATEGORY_3" contains "Dips & Salsa".

SQL query:

```
SELECT  
    p.BRAND,  
    SUM(t.FINAL_SALE) AS total_sales  
FROM Transactions t  
JOIN Products p ON t.BARCODE = p.BARCODE  
WHERE p.CATEGORY_3 = 'Dips & Salsa'  
GROUP BY p.BRAND  
ORDER BY total_sales DESC  
LIMIT 1;
```